Privacy Preserving Socio-Demographic Characteristics Identification From Smart Meter Data in Federated Setting

EEE 400(July 2025)- Final Examination
Date of Presentation: 9 June,2026

Presented By:

1.Md.Nasimul Islam Nayem
Id: 2006067

2.Moshiur Rahman
Id:2006099

Examination Committee:

1.Dr. Hafiz Imtiaz (Supervisor)
2.Dr. Hamidur Rahman (Member)
3.Dr. Maruf Ahmed(Member)

Outline of Presentation

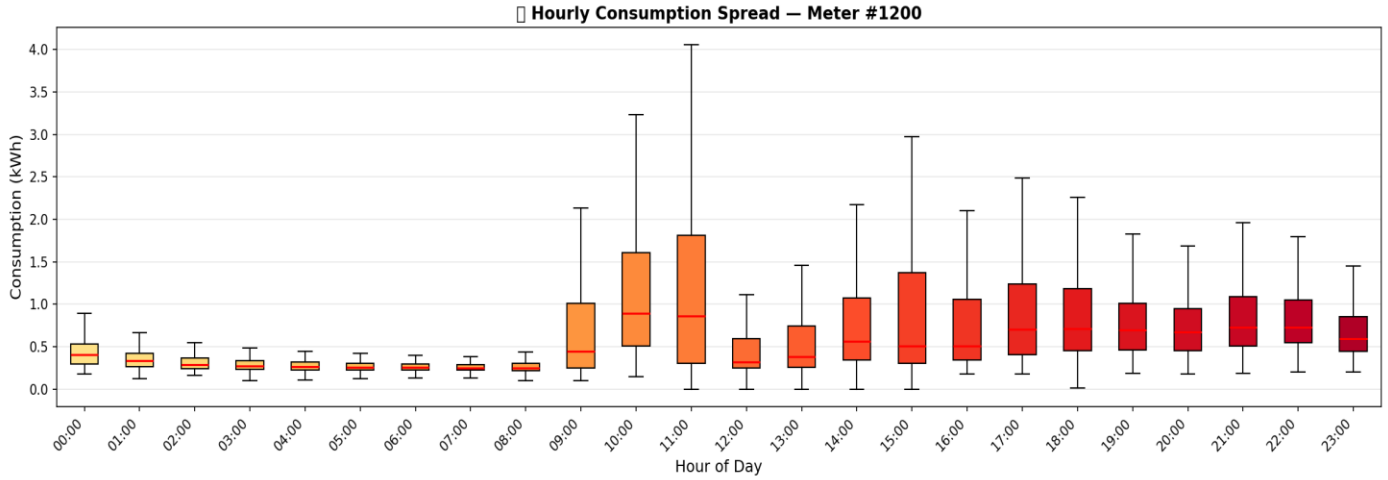
1. Summary
2. Background
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4. Novelty of our Work
5. Methodology and Design
6. Result and analysis
7. Limitation and Future work
8. Reference and Acknowledgement

1 Summary

Utility Companies collects huge amount of data and use that to train Machine Learning Models to offer diversified services by getting hidden insights of data, which raises a severe privacy issue. Despite the widespread use of **encryption systems**, the advancement of **quantum computing** poses a potential threat to their security. Therefore, we propose a novel framework to secure data without relying on encryption-based systems, utilizing smart meter data

2 Background

Raw Dataset: Irish Commission of Energy Regulation
Total Smart meter:6435
Household meter:4232, Rest are SME, industrial meter
Load data collection interval:30 mins (total 536 days)



1.Utility can perform unsupervised clustering, Load forecasting etc but can't predicts socio-economic condition of user.

2.If a utility offers lower peak-time tariffs for retired users, it must use supervised machine learning to predict consumer status.

3.Door to Door survey integrates socio-demographic characteristics(Job status, retirement status, floor size, home type etc) with load consumption data, but every one not willing to share.

4.A larger dataset improves model training performance but Strict rules on data privacy imposed by European General Data Protection Regulation (GDPR).

(Utility/retailer can't share data with each other and any third party operator)

Existing study, method, finding and gaps

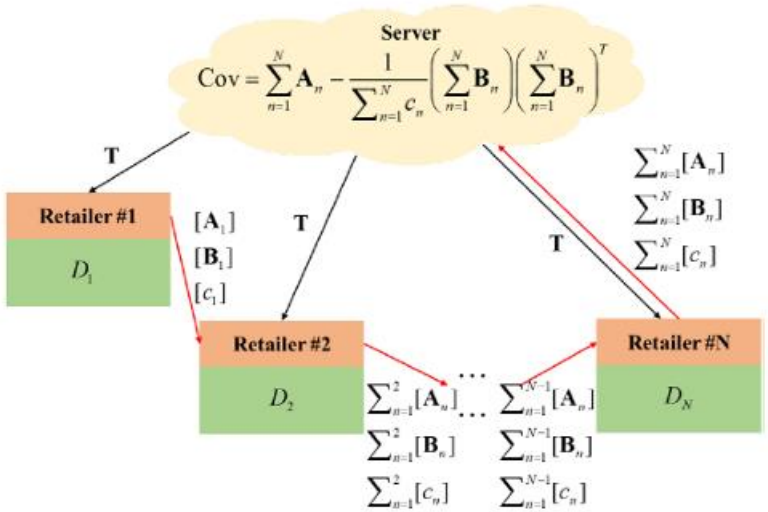


Fig. 3. Communication scheme of federated PCA.

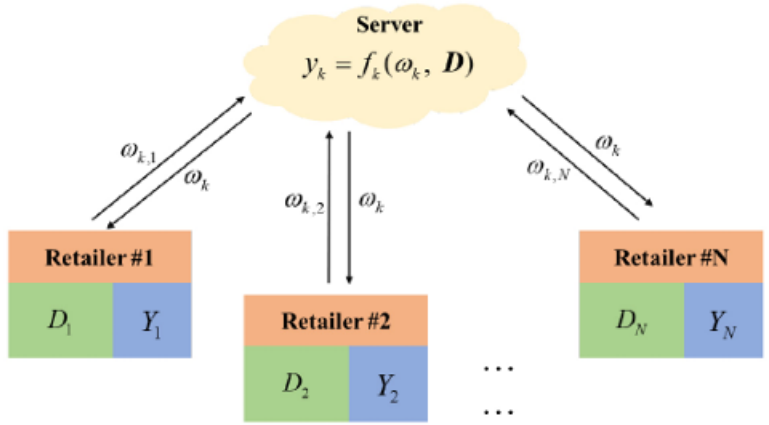


Fig. 4. Federated consumer characteristics identification in a competitive retail market.

D: input features
Y: Output Labeling

Retailer encrypts covariance matrix then combine with each other and sends to central server, which decrypts these summations!

4 Novelty of our work

Drawback of existing literature:

1.Central Server can be curious and may leak data
2.Homo-morphic encryption based pipelines vulnerable to quantum attacks, potentially exposing each user's data.

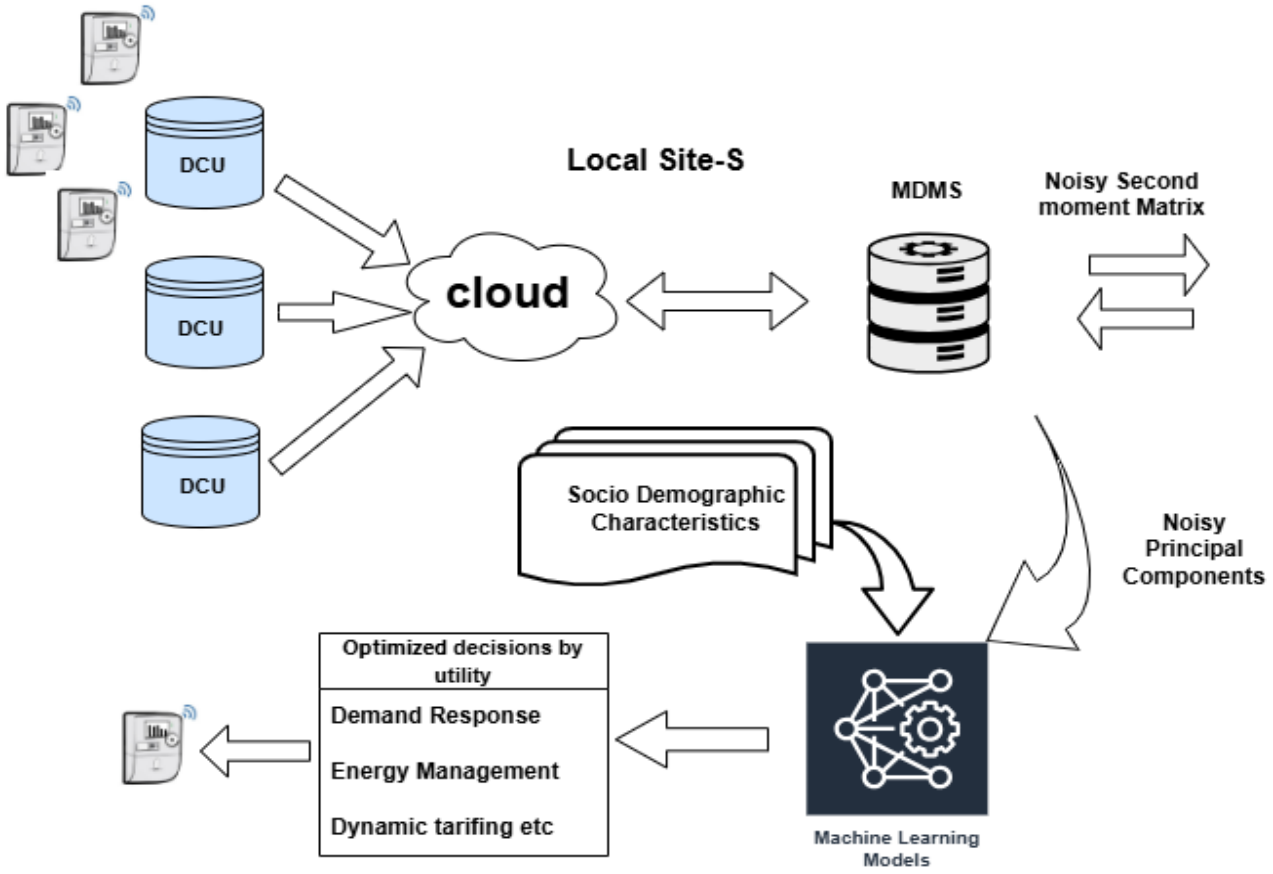
We propose a **Gaussian mechanism–based differentially private framework** that ensures **users' plausible deniability.**

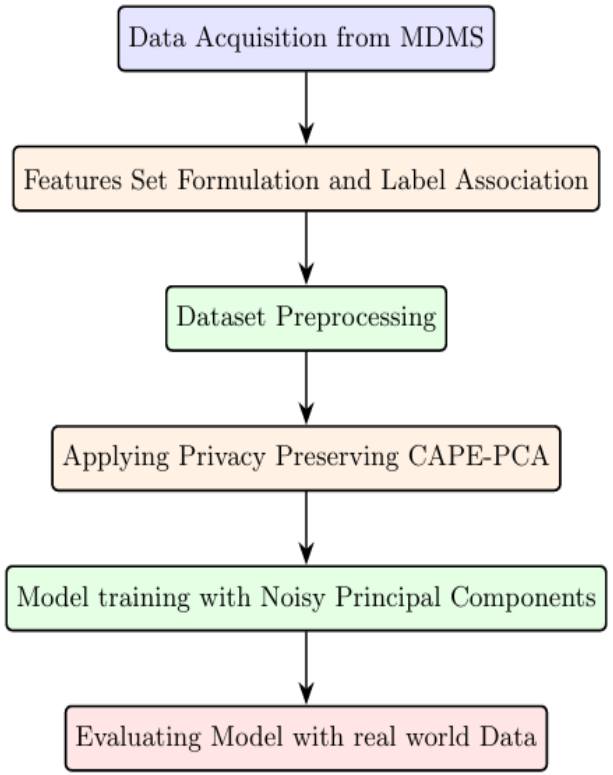
The algorithm provides (ε, δ)-differential privacy:

$$\boxed{\Pr[\mathcal{M}(D) \in \mathcal{O}] \leq e^{\varepsilon} \cdot \Pr[\mathcal{M}(D') \in \mathcal{O}] + \delta}$$

where D and D' are neighboring datasets differing in one sample.

Methodology &Design





Feature Engineering and Preprocessing

Features Set formulated according to existing Literature!

Total 71 features for complete load profiling.

Statistics: Basic load metrics (mean, max, min, total) for weekdays, weekends, and time periods (base, morning, midday, evening).

Ratios: Load factors & normalized comparisons (e.g., evening/midday, weekday/weekend ratios).

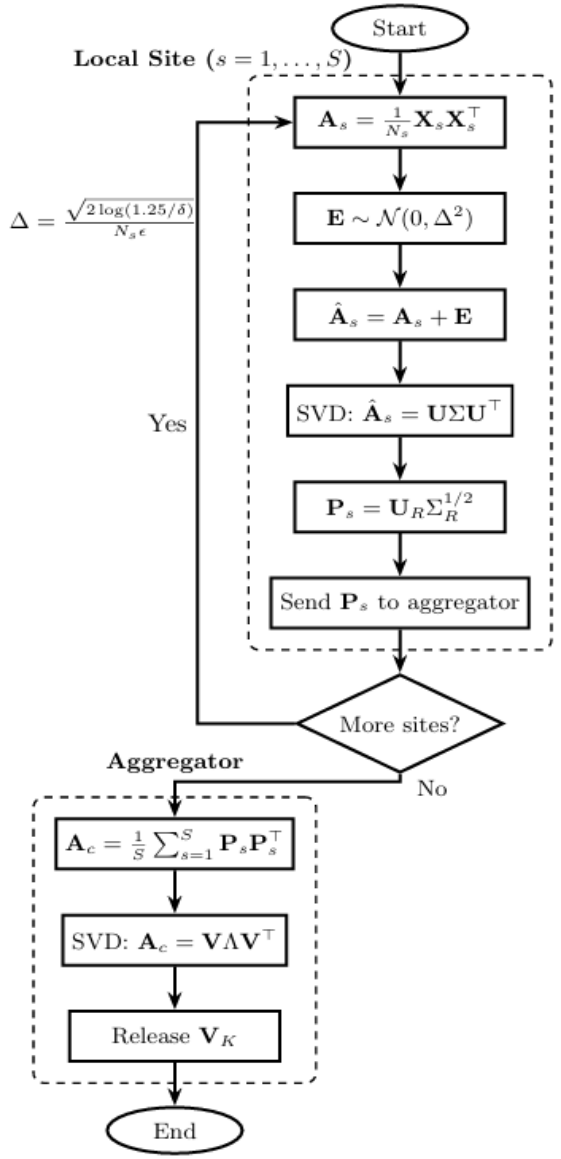
Distribution: Shape & behavior of load curves (quantiles, skewness, kurtosis, entropy, autocorrelation).

Preprocessing:

SMOTE: creates synthetic samples for the minority class by interpolating between existing minority class instances to balance imbalanced datasets.

1. **Mean centering**: $x_avg = \frac{1}{n} * \sum xi$; $\hat{x} = xi - x_avg$
2. **L2 norm** : $\|X\| < 1$, No single point can dominate the covariance matrix or principal components.

Privacy Preserving PCA for distributed setting



Standard deviation of Noise Matrix: $\Delta_{\epsilon,\delta} = \frac{1}{N_s\epsilon}\sqrt{2\log(\frac{1.25}{\delta})},$

Projected all data onto the top K principal components: Z = transpose(V_K). X_raw
V_K =Top K eigenvector

Z will be fed into the classifier

Dim(X_s)=D*N_s = 71*2028 (Site-2)
Dim(A_s) =D*D =71*71=Dim(Ac)=Dim(E)
Dim(U_R)=D*R(intermediate dimension) R>K
Dim(lemda)= D*D, eigen value of Ac
Dim(V_K)=D*K= 71*16
Dim(Z)=16*2028

Result and Analysis

1.Finding optimal PCs for best utility

Cumulative Variance is mostly used parameter to understand the percentage of total variance preserved by first K components. It helps to determine optimal number of PCs needed to retain a percentage of original dataset's information faithfully

Let $\lambda 1,\ldots,\lambda k$ be eigenvalues.

Total variance : $T=\sum_{j=1}^{k} \lambda j$

Cumulative variance explained by Top m PCs: $\frac{\sum_{k=1}^{m} \lambda k}{T}$

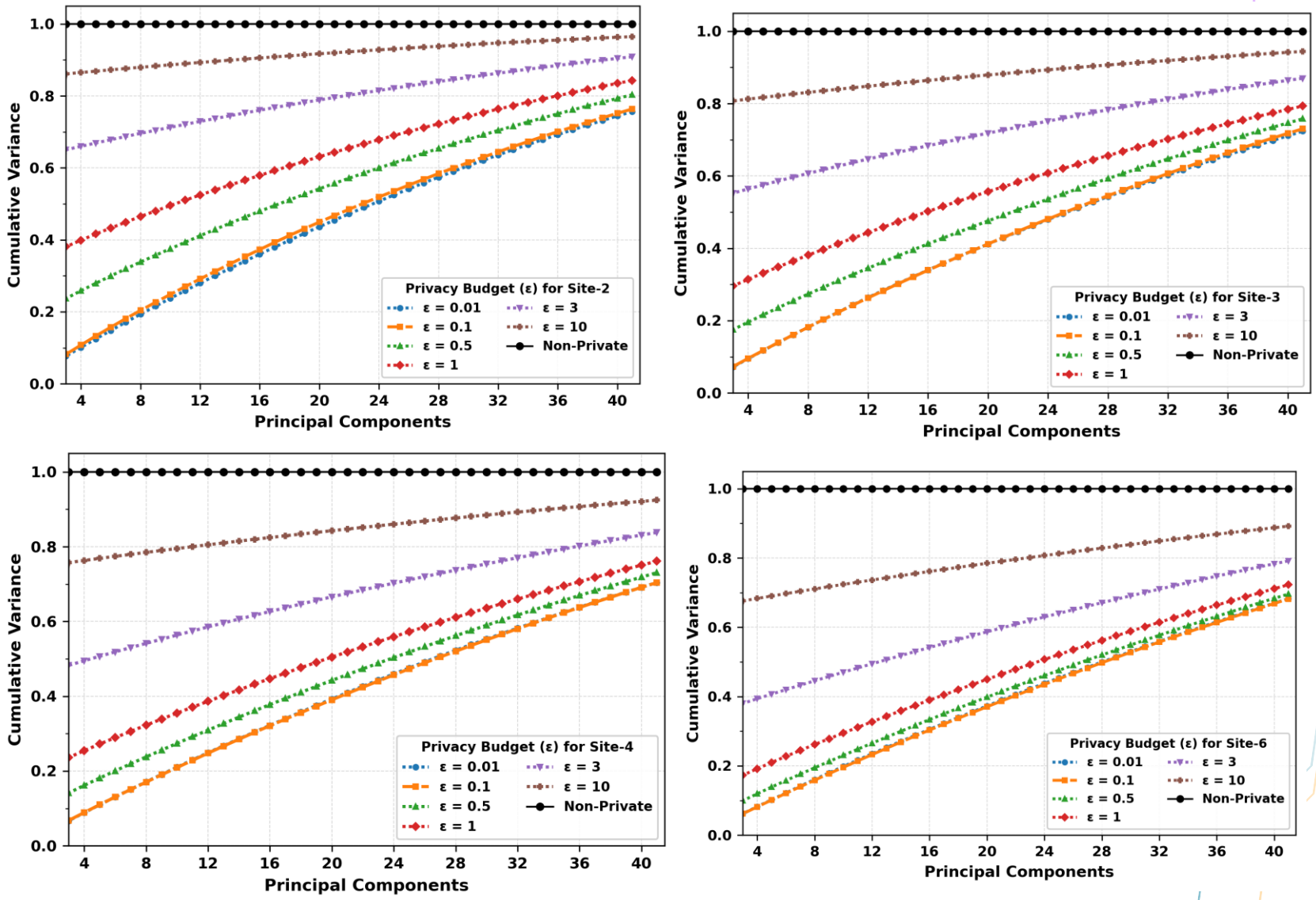
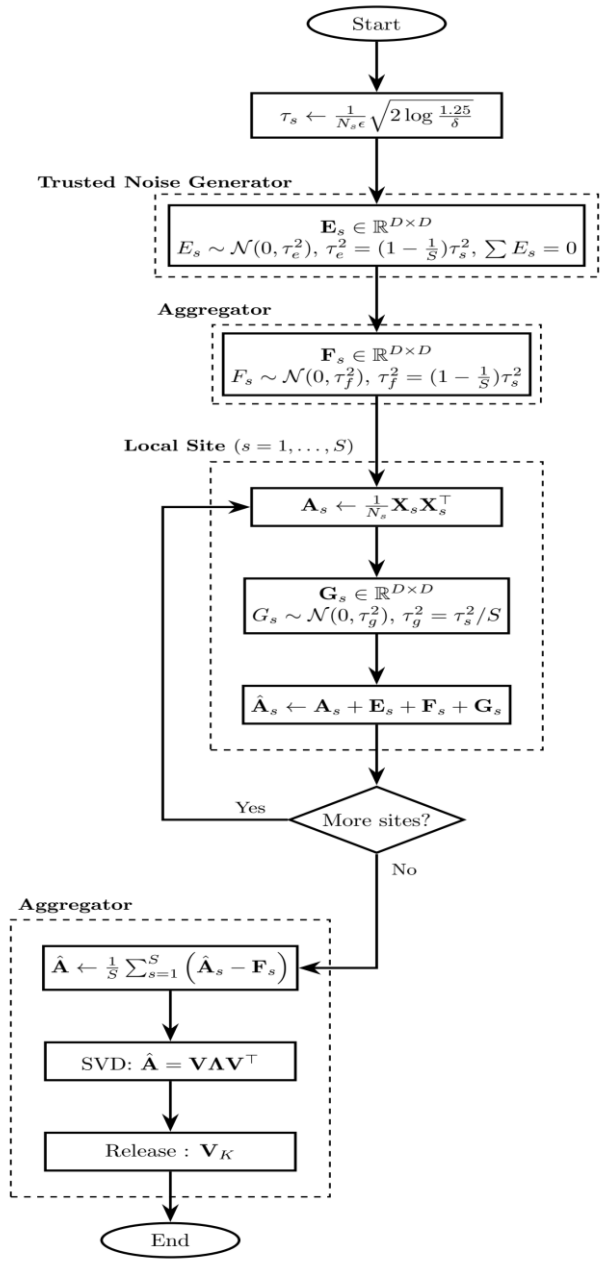


Fig: Comparison of CVE plots for different federated settings

CAPE PCA

- Fixed epsilon ϵ: fewer sites retain more information and require fewer PCs.
- Pooled data offers optimal utility but is infeasible due to privacy concern. CAPE-PCA addresses this limitation without sharing raw data.
- Aggregator-side cancellation of E_s, F_s, and G_s reduces injected noise.
- CAPE-PCA adds S^2-fold less noise than DP distPCA.
- Performance approaches that of centralized (pooled) PCA.



1.Finding optimal PCs for best utility for CAPE

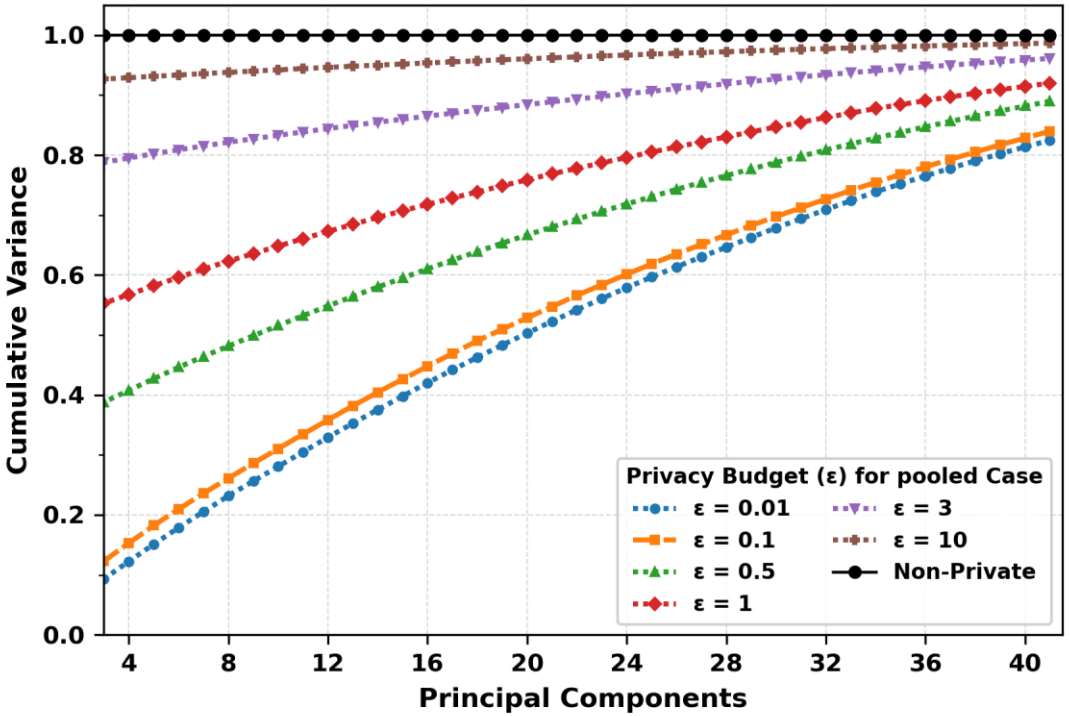


Fig: Comparison of CVE plots for Pooled /CAPE

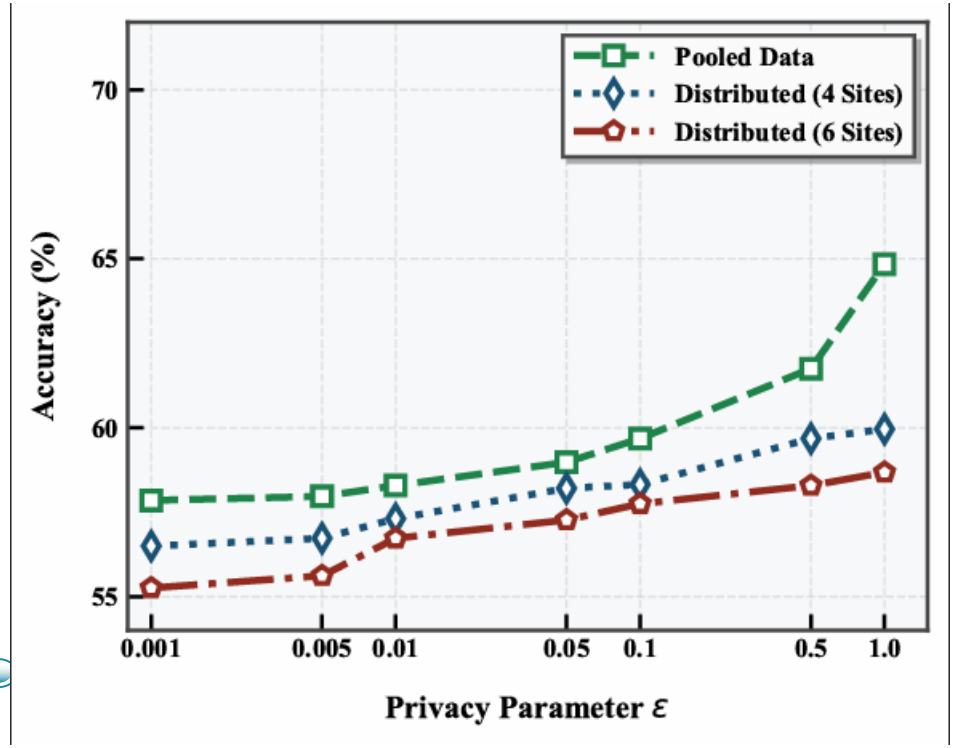
Table 3:Recommendation for CAPE

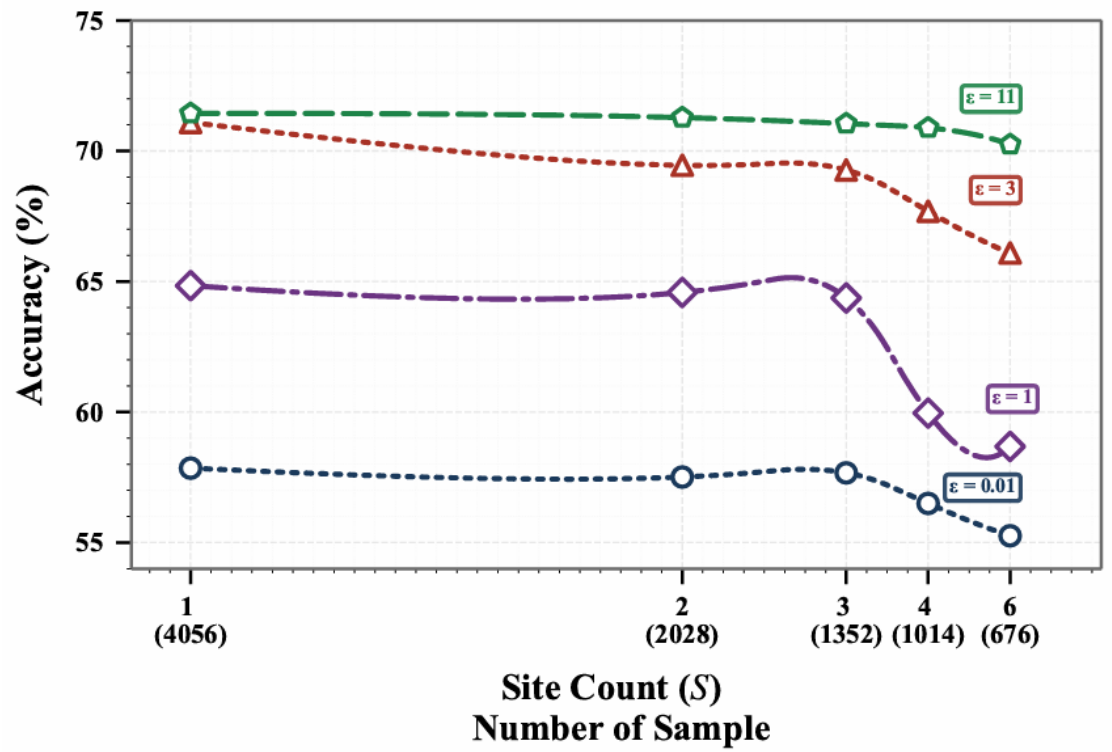
Privacy Parameter(ε)	*PCs Needed for ~90% Variance*	*Overall Recommendation in Pooled/ CAPA setting*
0.01	*>50*	*Higher Communication cost , Very strong Privacy & poor utility*
0.5	*>40*	*Higher Communication cost ,Strong Privacy & poor utility*
3	*~20*	*Moderate communication cost & privacy(optimal)*
10	*~3*	*Good for low-privacy needs, excellent utility*
Non Private	*~3*	*Best utility, no privacy*

Structure of CAPE: Distributed Setting
Performance of CAPE: Like Pooled Case

Performance Variation with Privacy Parameter epsilon & Number of site

Real world Test data must be supervised by projecting using V_k; otherwise, the accuracy remains around 50%.
This is because the classifier observes a different coordinate system when test data undergoes a noiseless dimensionality reduction.



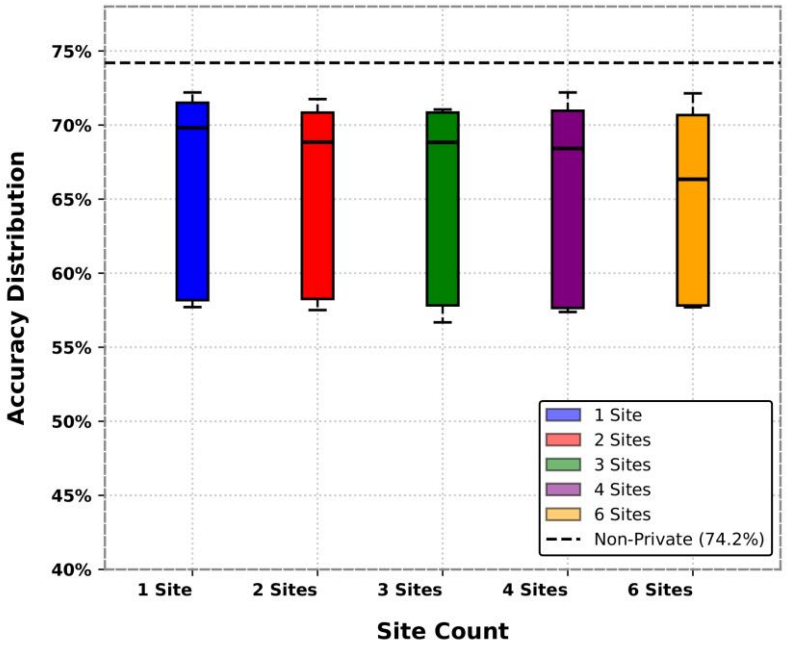


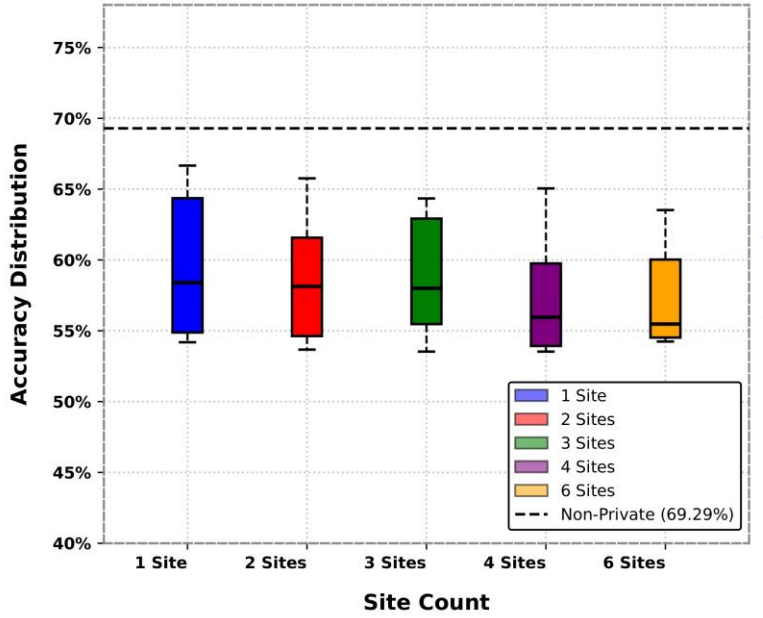
Higher epsilon → lower noise → higher classifier accuracy.
Pooled/ CAPE setting always outperforms other setting.

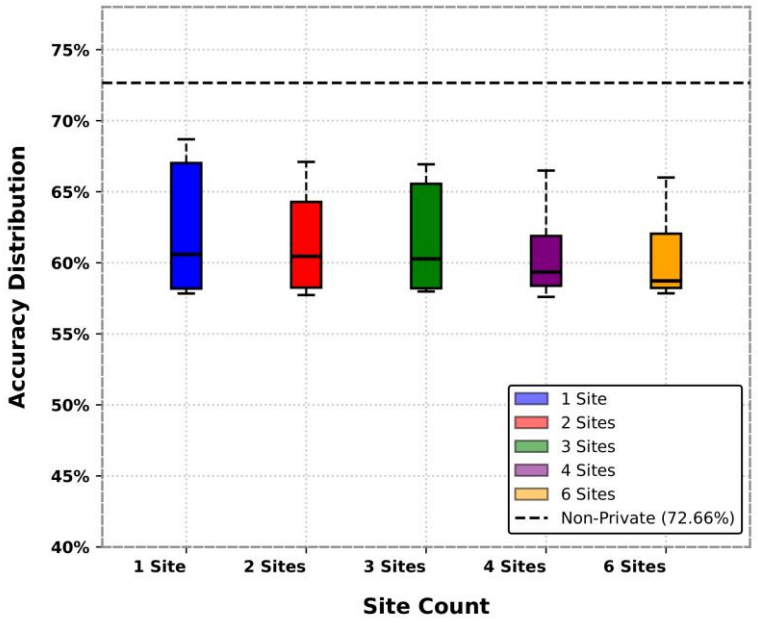
Performance Variation Model

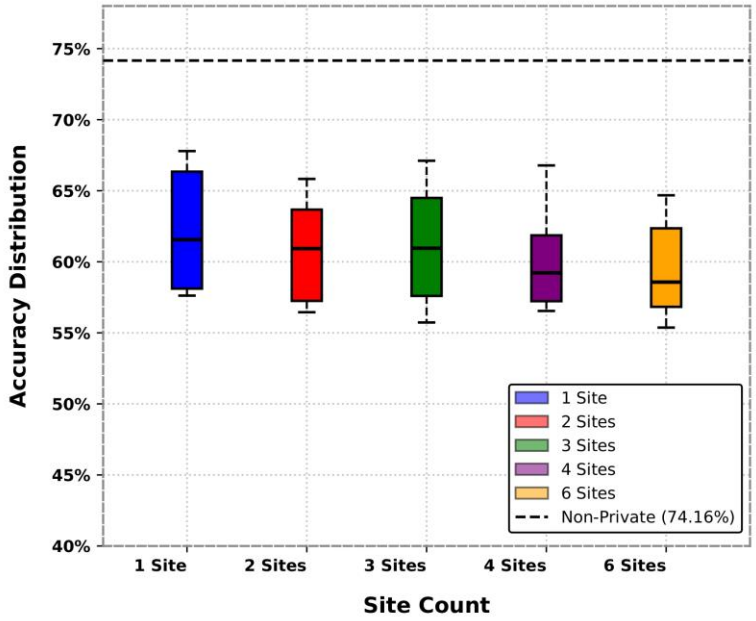
- Accuracy and IQR(Interquartile Range/Height) decreases with Number of site
- ANN is the most sensitive to noise
- XGBoost and SVM are the most resilient to site fragmentation; the median drop from S = 1 to S =6 is only about 2%.

Model performance comparison of ANN, Random Forest (RF), SVM, and XGBoost.









Limitation of Centralized training with privacy preserved features

Our main focus was to evaluate the feasibility of differential privacy in smart grid operations.

The proposed scheme focuses on privacy-preserving dimensionality reduction and training Classifiers with noise perturbed features.

While feeding Z into the classifier, we compute:

$$Z = \text{transpose}(V_K).\, X_pooled\,.$$

Notably, individual sites are not allowed to share or combine their raw data. It is the main drawback of centralized training

So, we plan to design another modified scheme to address this problem

Privacy Preserving Federated Learning

Local Classifiers will be trained with **Z_Local**

$$\text{Z_Local} = \text{transpose}(\text{V_K}).\,\text{X_Local}$$

Model update will be perturbed here again with noise then will be sent to central aggregator

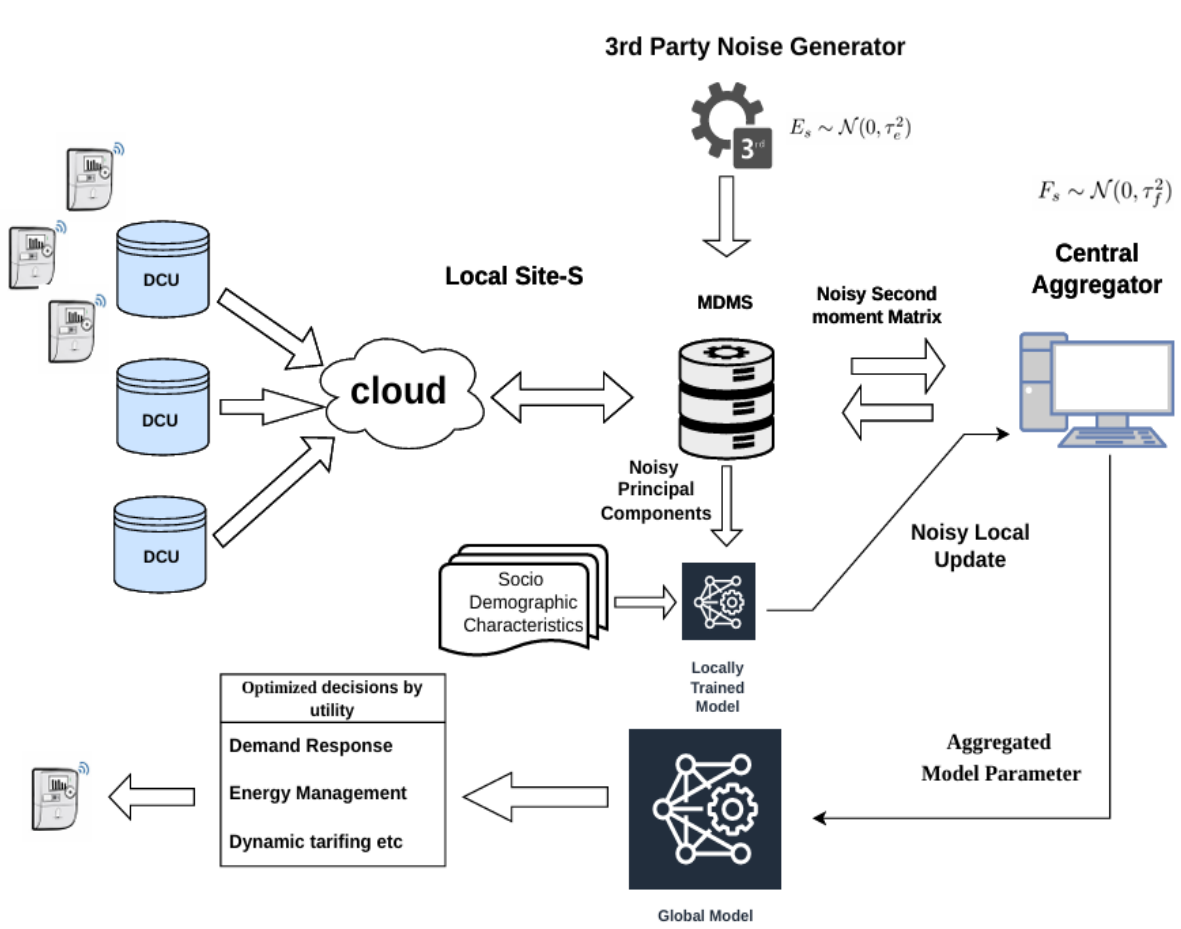
No risk of exposing Pooled raw dataset or Gradient descent

Recall that: Centralized Single Model is being trained with:

$$\text{Z_Pooled} = \text{transpose}(\text{V_K}).\,\text{X_pooled}.$$

1. Noise Multiplier: $\sigma = \dfrac{C\sqrt{2\ln(1.25/\delta)}}{\varepsilon}$
2. Per-Sample Clipping: $\bar{g}_i = \dfrac{g_i}{\max\left(1, \frac{\|g_i\|_2}{C}\right)}, \quad \|\bar{g}_i\|_2 \leq C$
3. DP-SGD Update: $\tilde{g} = \dfrac{1}{B}\left(\sum_{i=1}^{B} \bar{g}_i + \mathcal{N}(0, \sigma^2 C^2 I)\right)$
4. Parameter Update: $\theta_{t+1} = \theta_t - \eta \cdot \tilde{g}$
5. Federated Averaging: $\theta^{(r)} = \sum_{i=1}^{N} w_i \cdot \theta_i^{(r)}$
6. Sample-Based Weights: $w_i = \dfrac{n_i}{\sum_{j=1}^{N} n_j}$
7. Loss-Based Weights: $w_i = \dfrac{1/\ell_i}{\sum_{j=1}^{N} 1/\ell_j}$

Modified Framework

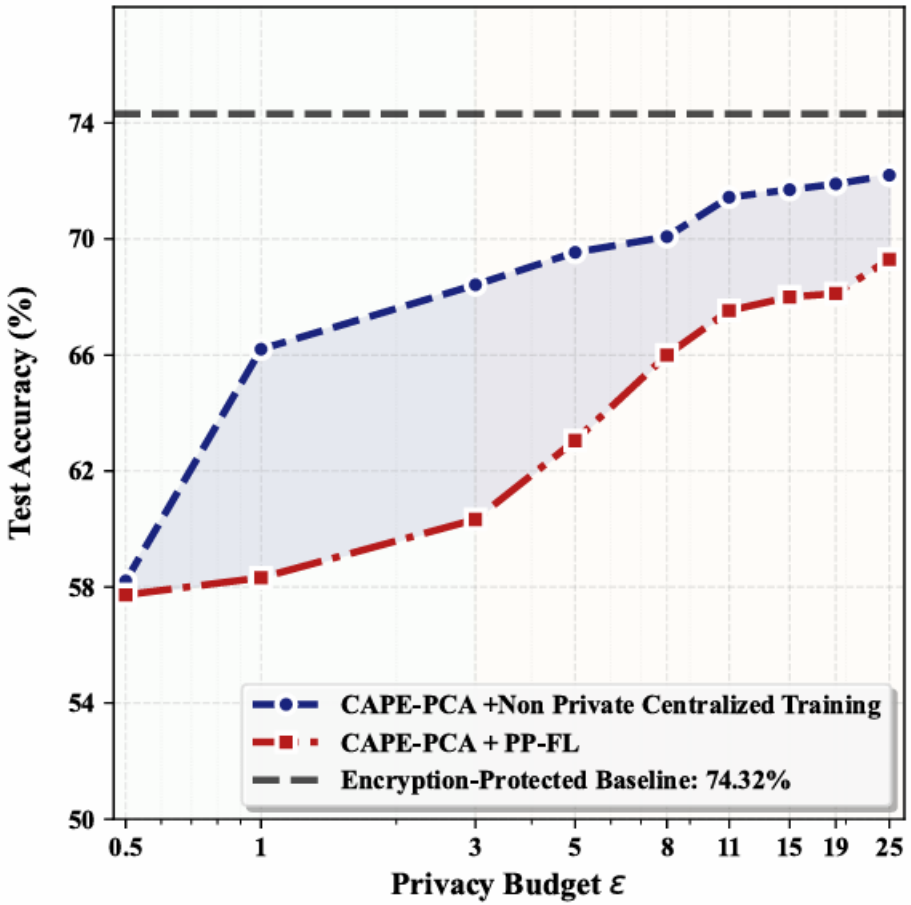


Result

Centralized training consistently performs better than federated learning by 3% to 8%.

Noise addition to model parameters when sending to central aggregator costs some accuracy

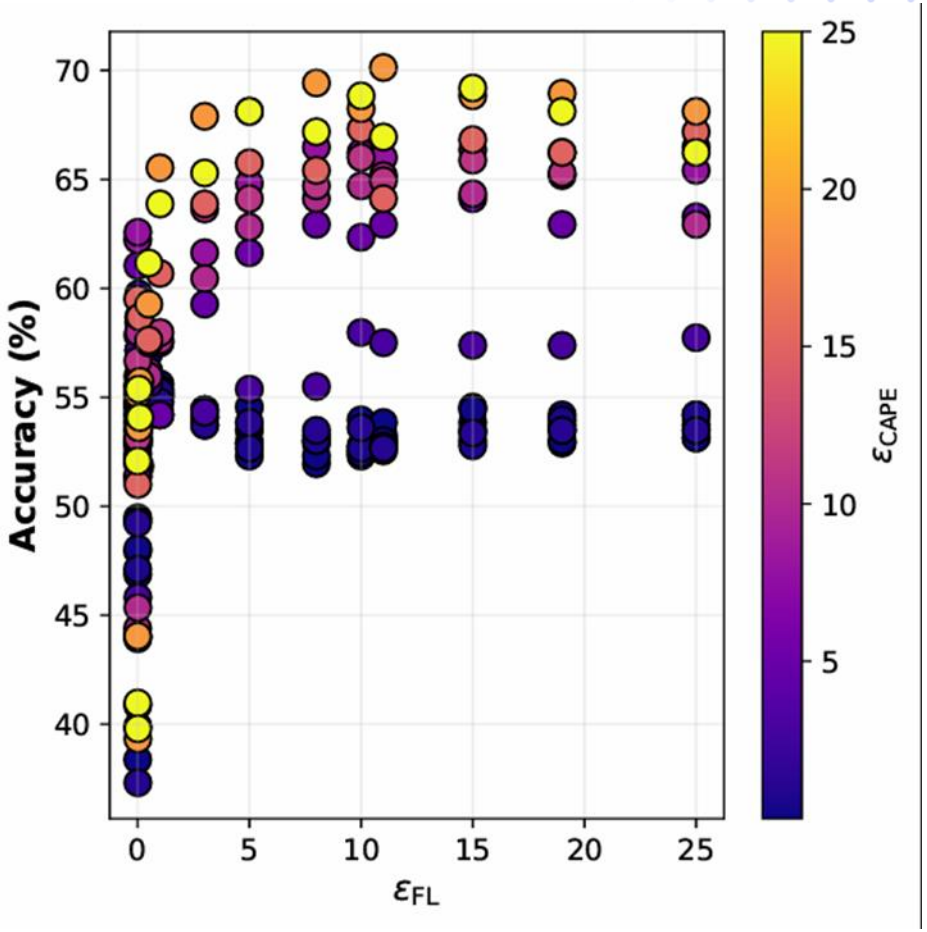
Higher epsilon → lower noise → higher classifier accuracy in both framework



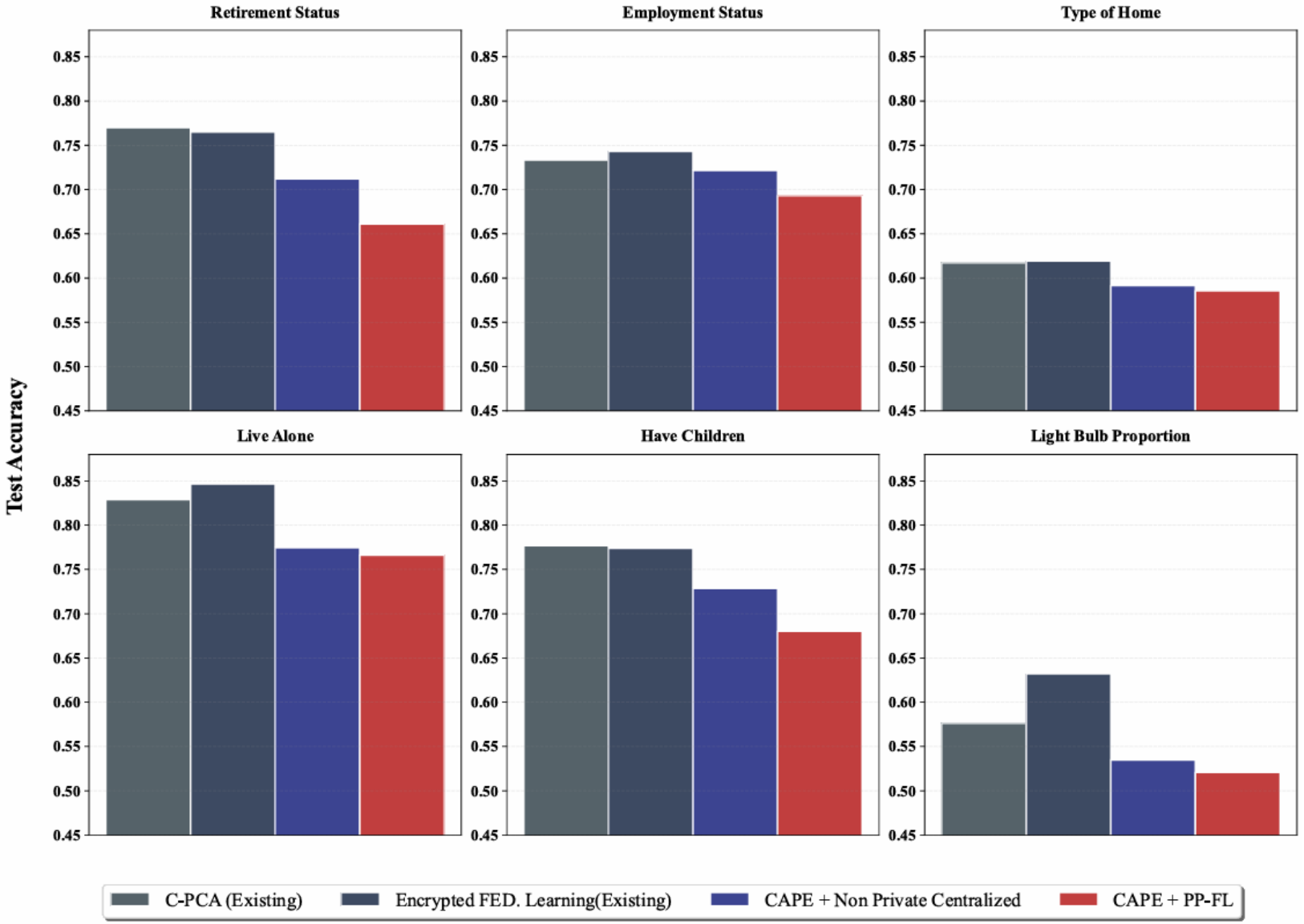
Optimal privacy budget selection

A minimum threshold of $\varepsilon \geq 3.0$ is recommended for both mechanisms to achieve acceptable performance (> 60%), while $\varepsilon \geq 11.0$ is recommended for high-accuracy applications (> 68%)

In industry such as Apple, Google use ($\varepsilon = 10$) to preserve the privacy



Comparison with existing Literature



References

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2. Hafiz Imtiaz and Anand D. Sarwate. Distributed differentially private algorithms for matrix and tensor factorization. IEEEJournal of Selected Topics in Signal Processing, 12(6):1449–1464, 2018.

3. I Wang, Imane Lahmam Bennani, Xiufeng Liu, Mingyang Sun, and Yao Zhou. Electricity consumer characteristics identi-fication: A federated learning approach. IEEE Transactions on Smart Grid, 12(4):3637–3647, 2021.

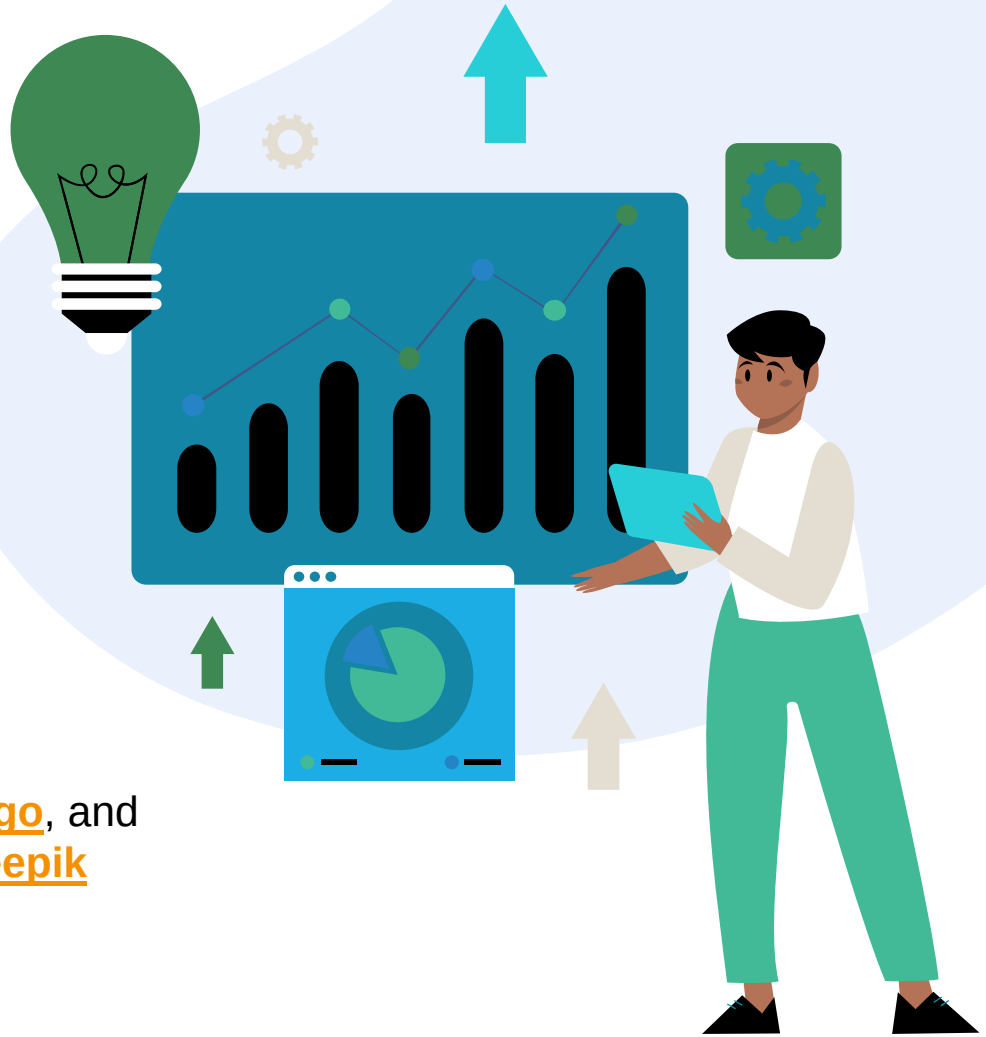
4. Abadi, M., Chu, A., Goodfellow, I., McMahan, H.B., Mironov, I., Talwar, K., Zhang, L.: Deep learning with differential privacy. In: Proceedings of the 2016 ACM SIGSAC Conference on Computer and Communications Security, pp. 308 318. ACM, ??? (2016)

Table 1: CO–PO–KPA Attainment Mapping with Smart Grid Thesis Evidence

CO No.	CO Statement	PO(s)	KPA	Evidence from Thesis
CO1	Literature review and problem identification	PO(b)	K1–K4, A1	Formulated the decentralized dimensionality mismatch problem in smart grids; audited limitations of homomorphic encryption against quantum attacks.
CO2	Research-based investigation and analysis	PO(d)	K8, A3	Evaluated algorithmic performance (ANN, SVM, XGBoost, RF) across multiple distributed site allocations ($S = 1$ to $S = 6$) under varying privacy budgets (ϵ).
CO3	Application of engineering knowledge	PO(a)	K1–K4, A2, A3	Implemented the mathematical matrix structures for CAPE-PCA and local L_2 sensitivity normalization bounding ($\|\|x\|\| \leq 1$) within a federated pipeline.
CO4	Professional and ethical conduct	PO(h)	K7, A3	Addressed individual data privacy compliance guidelines outlined under the European GDPR Article 89 and the California Consumer Privacy Act (CCPA).
CO5	Effective communication	PO(j)	A2	Detailed documentation of data alignment methods, mathematical proofs, and analytical trade-offs via clear writing and professional visual charts.
CO6	Contemporary engineering issues	PO(l)	K1–K6, A5, P6	Addressed data governance friction among competitive retailers while optimizing high-dimensional advanced metering infrastructure (AMI) datasets.

Thanks!

Do you have any questions?



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